AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
Credit Hours Engineering Programs – Bylaw 2018

Spring 2024	*Exam Date: 07.04.2024*	*Exam Time 60 min.*

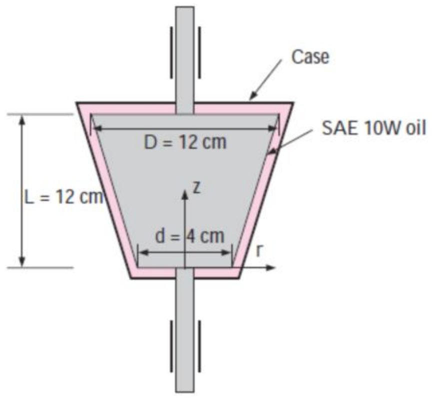
MEP 221s: Fluid Mechanics and Turbomachinery
MEP231: Fluid Dynamics
Midterm Examination

The Exam Consists of **Three** Questions in **Three** Pages — *Total Marks: 25 Marks*

Student Name: **ID:**

Question (1): (8 points)

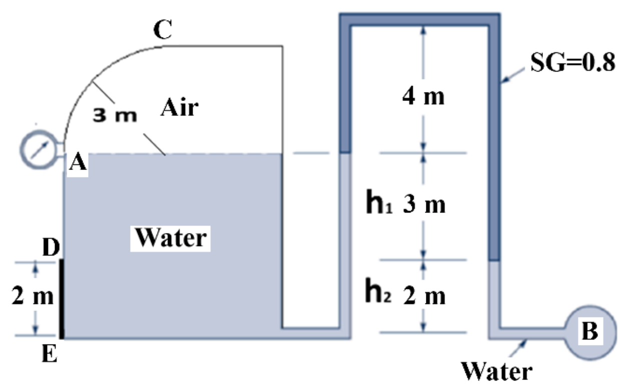
A frustum-shaped body is rotating at a constant angular speed of 200 rad/s in a container filled with SAE 10W oil at 20°C ($\mu = 0.1$ Pa.s), as shown in Figure. If the thickness of the oil film on all sides is 1.2 mm.



Determine:
(a) the power required to maintain this motion. **(5 points)**
(b) determine the reduction in the required power input when the oil temperature rises to 80°C ($\mu = 0.0078$ Pa.s). **(3 points)**

Question (2): (10 points)

A closed tank containing water is connected to an inverted U-tube manometer. A rectangular plate DE of 2 m height and 1 m width is placed at the bottom of the tank and a 1 m-long curved gate AC is located at the top left side of the tank as shown in the Figure. The liquid in the top part of the inverted U-tube manometer has a specific gravity of 0.8, and the remaining is filled with water. The upper part of the tank is filled with air. If the pressure at B is 205 kPa absolute and the atmospheric pressure is 1 bar.



Determine:

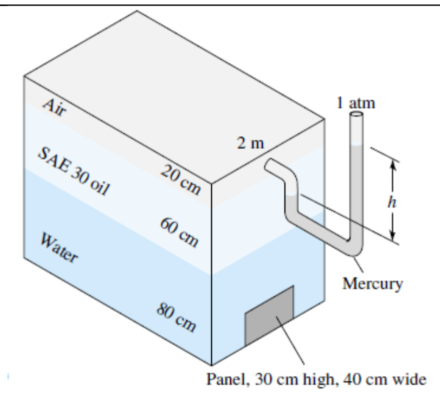
(a) The pressure gauge reading at A. **(2points)**

(b) The magnitude, direction, and location of the resultant hydrostatic force acting on the curved gate AC. **(2 points)**

(c) The magnitude, direction, and location of the resultant hydrostatic force acting on the plate DE (using the pressure prism method only). **(6 points)**

Question (3): (7 points)

For the closed tank in Figure, all fluids are at 20°C, and the airspace is pressurized. It is found that the net outward hydrostatic force on the 30-by 40-cm panel at the bottom of the water layer is 8450 N. For SAE 30 oil at 20°C ($\rho = 800\ \mathrm{kg/m^3}$).



Determine:

(a) The pressure in the airspace. **(4 points)**

(b) The reading (h) on the mercury manometer. **(3 points)**

Examination Committee:

Dr. Ashraf Ghorab - Dr. Walid Aboelsoud - Dr. Mohannad Emad